

Gradient Descent for Multiple Linear Regression

1. Equation is: $y = m_1 x_1 + m_2 x_2 + b$

We have,

$$x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 1

$$\begin{aligned} mx &= [(1)(-1) + (3)(2), (4)(-1) + (10)(2)] \\ &= [-1 + 6, -4 + 20] \\ &= [5, 16] \end{aligned}$$

Add bias vector b

$$\begin{aligned} \hat{y} &= mx + b \\ &= [5+1, 16+1] = [6, 17] \\ &= [6, 17] \end{aligned}$$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Step 2

Cost function

$$J(m, b) = \frac{1}{n} \sum (\hat{y} - y)^2$$

where,

$$\hat{y} = \sum m x + b$$

$$n = 2$$

Our formulas:

$$a) \frac{\partial J}{\partial m} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial m}$$

$$b) \frac{\partial J}{\partial b} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b}$$

$$c) \frac{\partial J}{\partial \hat{y}} = (-2/n) (\hat{y} - y)$$

$$d) \frac{\partial \hat{y}}{\partial b} = 1$$

$$e) \frac{\partial J}{\partial m} = 2/n \cdot x^T (\hat{y} - y)$$

$$f) \frac{\partial J}{\partial b} = (2/n) (\hat{y} - y)$$

Step 3

$$m_{\text{new}} = m_{\text{old}} - \alpha \cdot \partial J / \partial m$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \cdot \partial J / \partial b$$

Learning rate used

$$\alpha = 0.01$$

Predicted values from step 1

$$mx = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

$$MSE = (1^2 + 11^2) / 2$$

$$= \frac{102}{2}$$

$$= 51.0$$

$$\hat{y} = mx + b = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$$e = \hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Gradient wrt m = using $\partial J / \partial m$
 $= \frac{2}{n} X^T e$

$$X^T = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$X^T e =$$

$$X^T e = [(1)(1) + (4)(11), (3)(1) + (10)(11)]$$

$$= [1 + 44, 3 + 110]$$

$$= \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\partial J / \partial m = (2/2) \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

Gradient wrt b

$$\text{using } \frac{\partial J}{\partial b} = \frac{2}{n} e$$

$$= \frac{2}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1 - 0.45 \\ 2 - 1.13 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 - 0.01 \\ 1 - 0.11 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

Iteration 1.

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \quad b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$m^T x = \begin{bmatrix} (1)(-1.45) + (3)(0.87) \\ (-2)(-1.45) + (10)(0.87) \end{bmatrix} = \begin{bmatrix} 1.16 \\ 9.90 \end{bmatrix}$$

$$\text{Error}(e) = y - \hat{y} = \begin{bmatrix} 5 - 1.16 \\ 6 - 9.90 \end{bmatrix} = \begin{bmatrix} 3.84 \\ -3.90 \end{bmatrix}$$

$$\text{MSE} = \frac{1}{n} \sum e^2 = \frac{(3.84)^2 + (-3.90)^2}{2} = 6.5033$$

Gradient Δf

$$\frac{\partial J}{\partial m} = \frac{2}{2} x$$

$$x^T e = \begin{bmatrix} (1)(3.84) + (3)(-3.90) \\ (-2)(3.84) + (10)(-3.90) \end{bmatrix} = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = \frac{2}{2} \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

Gradient with respect to b

$$\frac{\partial J}{\partial b} = \frac{2}{2} \begin{bmatrix} 3.84 \\ -3.90 \end{bmatrix} = \begin{bmatrix} 3.84 \\ -3.90 \end{bmatrix}$$

New parameter update.

$$m_{\text{new}} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix} \quad b_{\text{new}} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.01 \begin{bmatrix} 3.84 \\ -3.90 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{bmatrix} = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} \quad = \begin{bmatrix} 0.99 + 0.0384 \\ 0.89 + 0.0390 \end{bmatrix} = \begin{bmatrix} 1.0284 \\ 0.9290 \end{bmatrix}$$

Iteration 2 results.

$$m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} \quad b = \begin{bmatrix} 1.0284 \\ 0.9290 \end{bmatrix}$$

Iteration 3

$$m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} \quad b = \begin{bmatrix} 1.0284 \\ 0.9290 \end{bmatrix}$$

$$\text{Error}(e) = \begin{bmatrix} 3.2149 - 5 \\ 7.3447 - 6 \end{bmatrix} = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

$$\text{MSE} = \frac{(-1.7851)^2 + (1.3447)^2}{2} \approx 2.4974$$

Gradient with respect to m.

$$X^T e = \begin{bmatrix} (1)(-1.7851) + (4)(1.3447) \\ (3)(-1.7851) + (10)(1.3447) \end{bmatrix} = \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix}$$

$$\therefore \frac{\partial \bar{y}}{\partial m} = \frac{1}{2} \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix} = \begin{bmatrix} 1.79685 \\ 4.04585 \end{bmatrix}$$

With respect to b

$$\frac{\partial \bar{y}}{\partial b} = \frac{1}{2} \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} = \begin{bmatrix} -0.89255 \\ 0.67235 \end{bmatrix}$$

Parameter updates

$$m_{\text{new}} = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - 0.01 \begin{bmatrix} 1.79685 \\ 4.04585 \end{bmatrix} = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - 0.01 \begin{bmatrix} -0.89255 \\ 0.67235 \end{bmatrix} = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

Iteration 4

$$m = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} \quad b = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

Predicted values.

$$m\bar{x} = \begin{bmatrix} (1)(-1.369037) + (3)(1.095583) \\ (4)(-1.369037) + (10)(1.095583) \end{bmatrix} = \begin{bmatrix} 1.9177 \\ 5.4797 \end{bmatrix}$$

$$\hat{y} = m\bar{x} + b = \begin{bmatrix} 1.9177 + 1.0364 \\ 5.4797 + 0.8987 \end{bmatrix} = \begin{bmatrix} 2.9541 \\ 6.3783 \end{bmatrix}$$

$$\text{Error}(e) = \begin{bmatrix} 2.9541 - 5 \\ 6.3783 - 6 \end{bmatrix} = \begin{bmatrix} -2.0459 \\ 0.3783 \end{bmatrix}$$

$$MSE = \frac{((-2.0459)^2 + (0.3783)^2)}{2} = \frac{2.1645}{2} \approx 2.1645$$

Gradient with respect to m.

$$X^T e = \begin{bmatrix} (1)(-2.0459) + (4)(0.3783) \\ (3)(-2.0459) + (10)(0.3783) \end{bmatrix} = \begin{bmatrix} -0.5326 \\ -2.3545 \end{bmatrix}$$

$$\therefore \frac{\partial \bar{y}}{\partial m} = \frac{1}{2} \begin{bmatrix} -0.5326 \\ -2.3545 \end{bmatrix} = \begin{bmatrix} -0.2663 \\ -1.17725 \end{bmatrix}$$

with respect to b:

$$\frac{\partial \bar{y}}{\partial b} = \frac{1}{2} \begin{bmatrix} -2.0459 \\ 0.3783 \end{bmatrix} = \begin{bmatrix} -1.02295 \\ 0.18915 \end{bmatrix}$$

Parameter updates

$$m_{new} = \begin{bmatrix} -1.3690 \\ 1.0756 \end{bmatrix} - 0.01 \begin{bmatrix} -0.5326 \\ 2.3548 \end{bmatrix} = \begin{bmatrix} -1.3690 + 0.0053 \\ 1.0756 + 0.0235 \end{bmatrix} \\ = \begin{bmatrix} -1.3637 \\ 1.1191 \end{bmatrix}$$

$$b_{new} = \begin{bmatrix} 1.0364 \\ 0.8787 \end{bmatrix} - 0.01 \begin{bmatrix} 2.0459 \\ 0.8783 \end{bmatrix} = \begin{bmatrix} 1.0364 - 0.0205 \\ 0.8787 - 0.0088 \end{bmatrix} \\ = \begin{bmatrix} 1.0568 \\ 0.8749 \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1.3637 \\ 1.1191 \end{bmatrix}$$

$$b_{new} = \begin{bmatrix} 1.0568 \\ 0.8749 \end{bmatrix}$$